Paolo Alberto Mongini

Curriculum Vitae

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About me

I am a PhD student in Electronic, Computer Science, and Electrical Engineering at the University of Pavia, where I have been conducting research since 2021. My work focuses on the development, simulation, and testing of new techniques to improve diabetes management for patients treated with artificial pancreas.
Starting from glucose sensor readings, I work with my research group on the development of fault detection methods, machine learning models, and control techniques aimed at keeping patients' glycemia within a safe range by acting on insulin pumps in a closed-loop framework.
Furthermore, I have the opportunity to mentor students on their Master's projects, deliver seminars and lectures, and present my research at national and international conferences.

I am currently undertaking a research period at EPFL, where I am working on the development of safe-by-design machine learning models to predict patients' future glycemia.

Previously, I obtained my Master's degree in Industrial Automation and my Bachelor's degree in Industrial Engineering (Energy). My academic background includes signal processing, control systems, system identification, statistics, machine learning, electronics, and electrical machines.

In my free time, I enjoy reading, listening to podcasts, and going for walks in nature.

Current Position

Oct2025–Present **Visiting Researcher**, *École Polytechnique Fédérale de Lausanne/EPFL*, Visiting period during my Ph.D. at the Dependable Control and Decision Group.
Research areas: Machine Learning model identification with stability and safety guarantees by design.

2023–Present **PhD Student in Electronic, Computer Science and Electrical Engineering**, *University of Pavia*, Identification and Control of Dynamic Systems Laboratory.
Research areas: development of fault detection, signal processing, system identification, control design, and machine learning techniques to improve the performance and safety of artificial pancreas systems for diabetes patients.

Experience

2021-Present **Tutor of Automatic Control, Physics, Electrotechnology and System identification**, *University of Pavia*, Pavia, Italy
Tutor for bachelor's degrees in *Electronic and Computer Engineering*, *Bioengineering*, and *Industrial Engineering*

Education

2021–2023 **Master Degree**, *University of Pavia, Italy*, Industrial automation engineering, *110/110 cum Laude*.
During my master's program, I completed coursework in advanced control techniques for industrial processes and electric motors, as well as statistical modeling and machine learning.

Master Thesis

Title *Controller design and recipe optimization for an automated perfume ingredient dispenser*

Abstract The thesis investigates the model identification and control and real-world implementation of an industrial system. A simulated environment was developed to recreate a flow control problem, where the controller was designed and validated. The solution was then implemented on real hardware in collaboration with a machinery manufacturing company for the cosmetics industry, involving PLC programming, inverter, sensor and actuator integration. The deployed system exhibited real-world behavior closely matching the simulation results, confirming effectiveness of the simulation-based design approach.

2018–2021 **Bachelor Degree**, *University of Pavia*, Italy, Industrial engineering - energy
106/110

Thesis

Title *Realization and validation of the dynamic model of a double inverted pendulum.*

Abstract The Thesis explores the realization and validation of the dynamic model of a rotating double inverted pendulum. A three-dimensional state-space model is developed to simulate the system dynamics and analyze its behavior. In addition, the work focuses on the setup and integration of the hardware components, including the power supply, H-bridge motor driver, sensors, and wiring. The system is interfaced with Arduino and Raspberry Pi to enable model validation and future controller integration.

Languages

Italian Mother tongue
English C2
Chinese A2

Technical skills

Programming Languages: Matlab, Simulink, Julia, C, Python, ST

Graphics: illustration (Affinity), 3D modeling (Inventor)

Machine Learning: regression and classification models (SVM, MLP, neural networks), computer vision (CNNs), time-series modelling (LSTM, REN, SSM), Large Language Models (Transformer), machine learning libraries (Pytorch, TensorBoard, Optuna)

Data Exploration: data processing (Pandas, Numpy) visualization (Matplotlib)

Productivity: Microsoft Office Suite, Google Workspace, Open Office, LATEX

Versioning: Git, GitHub, GitKraken